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12.7. HONEYCOMB AND FOAMS VERSUS FREQUENCY

Table 12.7 shows the parametric fits to permittivity for honeycomb and foam composites. Square test samples were cut from larger production pieces. Test samples were square shapes greater than 30 cm on each side. Thicknesses were typically 2.5 cm. Samples were measured with Fabry–Perot and/or focused beam system. The functional forms are shown at the beginning of the table. Functions are those causal functions that have been derived in Chaps. 2 and 3 for dielectric and magnetic media. Other data are shown when available. These include density, measured frequency span serving as basis for fits, tabulated information, and chemical and/or physical composition. One foam (**Fig. 12.10**, top) and one honeycomb (**Fig. 12.10**, bottom) are shown. In the tables exponentials are often shown as bE—which means $b \cdot 10^{-a}$. Thus, $-2\text{E}-004-7.6\text{E}-003j$ is equal to $-0.0002-j0.0076$. Nominal data from web-accessible external databases (e.g., Le Reseau Composites Network) are shown when available.

Table 12.7 *Foams/Honeycombs*

Sample ID							
0 or 90 degree rotations of the square shape sample (38 × 38 × 1 cm)							
$\epsilon(f) = B + 2Cf^D + G\{1 - J * (f - H)^2 - j2If\}^{-1}$							
	B	C	D	G	H	I	J
Syntactic foam F6555 (20–100 GHz) (nominal $\epsilon_r = 1.9$, $\tan\delta = .005$, Le Reseau Composites Network, icomposites.com)							
0 degree	0.52– 3E-003j	0.5– 1E-002j	–0.21+ 1E-002j	0.63– 3E-003j	2E-003– 3E-004j	–3E-003j	–2E-005– 4E-006j
90 degree	0.63+ 4E-003j	0.75+ 2E-002j	–0.14– 1E-002j	0.6– 1E-004j	1E-003	4E-004– 2E-003j	–2E-005– 4E-006j
Honeycomb HRH-310 meta-aramid fiber polyimide, 1/8-in. cell (10–60 GHz) (nominal $\epsilon_r = 1.015$ –1.043, $\tan\delta = .0002$ –.0018)							
0 degree	0.264– 4E-004j	0.266– 4E-004j	–3E- 002+ 1E-002j	0.263– 7E-004j	2E-003+ 1E-004j	–8E- 004– 3E-003j	–7E- 005+ 1E-005j
90 degree	0.254+ 2E-004j	0.254+ 2E-004j	2E-004	0.254+ 2E-004j	–2E-006	–2E- 005– 7E-004j	–2E- 005+ 1E-006j

Sample ID

0 or 90 degree rotations of the square shape sample (38 × 38 × 1 cm)

$$\epsilon(f) = B + 2Cf^D + G\{1 - J*(f - H)^2 - j2If\}^{-1}$$

B

C

D

G

H

I

J

Honeycomb HRP ¼-in. cell (30–60 GHz) fiberglass reinforced phenolic (nominal $\epsilon_r = 1.09$, $\tan\delta = .003$, CEMENTACH Group)

0 degree	0.331+ 4.5E-002j	0.378+ 3.5E-002j	–0.111+ 9E-002j	0.634+ 2E-004j	1.8E-002+ 2E-002j	–2E-002+ 5E-003j	1E-004+ 5E-004j
90 degree	0.33+ 9.8E-002j	0.193+ 4E-002j	0.24– 7E-002j	1.2E-002 –1.5E-002j	0.13– 6E-002j	6E-004– 3E-002j	–9E-004+ 9E-006j

Honeycomb HRP 1/16-in. cell (30–60 GHz) fiberglass reinforced phenolic

0 degree	0.297+ 4E-003j	0.297+ 4E-003j	–1E-003+ 2E-003j	0.31+ 5E-003j	7E-003+ 8E-004j	–3E-003+ 9E-003j	2E-004+ 1E-004j
90 degree	0.267+ 6E-003j	0.267+ 6E-003j	4E-004+ 1E-004j	0.267+ 6E-003j	4E-004+ 1E-004j	–1E-004– 9E-004j	–4E-005– 5E-006j

Honeycomb HRH-78 meta-aramid ¼-in. cell (18–60 GHz)

0 degree	0.258+ 7E-004j	0.259+ 7E-004j	2.3E-002 +7E-004j	0.258+ 6E-004j	–1.3E-003– 1E-004j	–1E-004– 1E-003j	–4.6E-005+ 2E-006j
90 degree	0.27– E-004j	0.27– 2E-004j	–6E-003+ 2E-004j	0.27– 4E-004j	1E-003– 5E-004j	1E-004+ 1.5E-003j	6E-005– 3E-006j

Honeycomb HRH-10 1/8-in. cell, meta-aramid fiber phenolic (18–60 GHz)

0 degree	0.258+ 7E-004j	0.259+ 7E-004j	2.5E-002+ 7E-004j	0.258+ 6E-004j	–1.3E-003– 1E-004j	–1E-004– 1E-003j	–5E-005+ 2E-006j
90 degree	0.253+ 3E-004j	0.255+ 3E-004j	2.9E-002+ 4E-004j	0.251+ 3E-004j	–1.5E-003 003	–2E-005– 5E-004j	–4E-005+ 3E-007j

Honeycomb para-aramid fiber [–CO–C₆H₄–CO–NH–C₆H₄–NH–]_n, HRH-49 para (5–40 GHz) (nominal $\epsilon_r = 1.03$ –1.07, $\tan\delta = .001$ –.003, Le Reseau Composites Network)

Sample ID

0 or 90 degree rotations of the square shape sample (38 × 38 × 1 cm)

$$\epsilon(f) = B + 2Cf^D + G\{1 - J * (f - H)^2 - j2If\}^{-1}$$

B C D G H I J

0 degree	0.266+ 6E-004j	0.266+ 6E-004j	2E-003	0.267+ 6E-004j	4E-004	2E-003j	1E-004
90 degree	0.262+ 4E-004j	0.263+ 4E-004j	-1.7E- 002	0.262+ 4E-004j	-5E-004	-2E-003j	-3E-005
Honeycomb HRH-327 fiberglass reinforced polyimide 3/16 in. (30–60 GHz)							
0 degree	0.267+ 6E-003j	0.267+ 6E-003j	4E-004 +1E- 004j	0.267+ 6E-003j	4E-004– 1E-004j	-1E-004– 9E-004j	-4E-005 +4E- 006j
90 degree	0.297+ 5E-003j	0.297+ 5E-003j	-1E-003 +2E- 003j	0.31+ 5E-003j	7E-003 +8E- 004j	-3E-003 +9E- 003j	2E-004 +1E-004j
Honeycomb ES PEI-E-glass, 5 lb/ft ³ (18–60 GHz)							
0 degree	0.253– 1E-003j	0.257– 7E-004j	4.2E- 002 +1.1E- 002j	0.254– 2E-003j	-2E- 003– 1E-003j	-2E- 003+ 7E-004j	8E-007+ 3E-005j
90 degree	0.271– 2E-004j	0.271– 2E-004j	-7E- 003– 2E-003j	0.272+ 1E-004j	-3E- 003+ 2E-003j	2E-004– 4E-003j	-1E-004
PBI Honeycomb (2–100 GHz)							
0 degree incidence (4–100 GHz)	0.286+ 2E-003j	0.288+ 2E-003j	-4E- 002– 8E-003j	0.292+ 2E-003j	-3E- 003– 2E-004j	5E-004– 3E-003j	-9E- 006– 3E-006j
20 degree incidence (>70 GHz)	0.41– 0.12j	0.4– 0.25j	-0.14– 0.14j	0.297– 7E-002j	8E-003– 6E-003j	3E-003– 1E-002j	-1E-004– 4E-005j
40 degree incidence (>70 GHz)	0.52– 3E-002j	0.54– 2E-002j	8.3E- 002+ 1.9E- 002j	0.462+ 5E-002j	8E-002+ 4E-003j	3E-002– 7E-002j	-2E-003– 8E-004j
60 degree incidence (>70 GHz)	0.32– 1E-002j	0.32– 1E-002j	-3E- 002– 1E-002j	0.35+ 3E-003j	-2E- 003– 3E-003j	2E-003– 1E-003j	2E-005– 3E-005j

Sample ID

0 or 90 degree rotations of the square shape sample (38 × 38 × 1 cm)

$$\epsilon(f) = B + 2Cf^D + G\{1 - J*(f - H)^2 - j2If\}^{-1}$$

B

C

D

G

H

I

J

Honeycomb polycarbonate 1/8-in., 0.149 g/cc (8–26 GHz)

0 degree	0.293+ 3E-004j	0.293+ 3E-004j	8E- 003– 2E-003j	0.294+ 3E-004j	–1.4E- 003– 1.2E-003j	5E-004+ 3E-003j	6E-005+ 1E-005j
90 degree	0.304– 5E-003j	0.31– 5E-003j	–5E- 002+ 2E-002j	0.303– 4E-003j	2E-003– 8E-004j	–4E- 004– 3E-003j	–5E- 005+ 6E-006j

Honeycomb polyimide ¼-in. (18–45 GHz)

0 degree	0.288	0.288	–4E- 004	0.287– 2E-004j	4E-004– 3E-005j	2E-004+ 5E-003j	2E-004– 2E-006j
90 degree							

Polyurethane foam 10 lb/ft³ (18–100 GHz)

0 degree	0.293– 2E-004j	0.293– 2E-004j	7E- 003+ 2E-003j	0.291– 1E-004j	–3E- 004– 1E-004j	2E-005- 5j	–1E-005– 9E-007j
90 degree							

Polyurethane foam 6 lb/ft³ (18–100 GHz)

0 degree	0.278+ 4E-004j	0.278+ 4E-004j	1E-004– 1E-004j	0.278+ 4E-004j	5E-007– 2E-007j	5E-005+ 2E-004j	3E-006– 6E-007j
90 degree							

Carbon reticulated foam (500 MHz to 6 GHz)

Sample C	1.55+ 0.28j	–0.64+ 1.81j	–0.95+ 0.16j	0.93+ 0.86j	0.5+ 0.46j	–0.41– 0.11j	–0.14– 5E-003j
Sample B	1.59+ 0.39j	–0.43+ 1.61j	–1.1+ 0.47j	–0.5+ 1.44j	0.31– 0.12j	–0.46– 0.11j	–0.11+ 5E-002j
Sample A	1.28– 0.55j	–0.32+ 2.61j	–0.84+ 0.11j	1.55+ 0.44j	–6E- 002+ 0.31j	–0.44+ 0.17j	–4E- 002+ 0.15j

Sample ID

0 or 90 degree rotations of the square shape sample (38 × 38 × 1 cm)

$$\epsilon(f) = B + 2Cf^D + G\{1 - J * (f - H)^2 - j2If\}^{-1}$$

B

C

D

G

H

I

J

110 Polymethacrylimide-1 foam (1–100 GHz) 6.9 lb/ft³

0.292+	0.292+	0	0.292+	0	0	0
2E-003j	2E-003j		2E-003j			

110 Polymethacrylimide-2 foam (1–100 GHz) 6.9 lb/ft³

0.321–	0.365–	–0.12+	0.318–	–1E-003+	–3E-004–	–2E-005+
3E-003j	3E-003j	1.4E-002j	4E-003j	2E-003j	2E-003j	4E-006j

200 Polymethacrylimide foam (18–100 GHz) 12.5 lb/ft³

0.322+	0.322+	7E-004+	0.322+	1E-004+	–5E-004+	2E-005+
3E-003j	3E-003j	2E-003j	3E-003j	1E-004j	8E-004j	8E-006j

300 Polymethacrylimide foam (18–60 GHz) 18.7 lb/ft³

0.273+	0.313+	0.12–	0.161–	3.3E-002+	–3E-002j	–2E-003
7E-003j	5E-003j	1E-003j	8E-003j	2E-003j		

51 Polymethacrylimide foam (8–26 GHz) 3.1 lb/ft³

0.289–	0.305–	–8.2E-002	0.287–	–1E-003+	–1E-004	–2E-005+
2E-003j	2E-003j	+7E-003j	2E-003j	3E-004j	–2E-003j	1E-006j

71 Polymethacrylimide foam (9–17 GHz) 4.4 lb/ft³

0.2753+	0.2753+	2E-003+	0.2754+	3.8E-003+	–1.2E-003+	–4E-004+
1.7E-003j	1.7E-003j	4E-004j	1.7E-003j	1E-003j	4.1E-003j	1E-004j

31 Polymethacrylimide foam (8–26 GHz) 1.9 lb/ft³

0.2625+	0.2625+	–5E-004	0.2625+	0	3E-004j	1E-004
2E-004j	2E-004j		2E-004j			

Polystyrene foam 7.8 lb/ft³ (9–18 GHz)

Sample ID

0 or 90 degree rotations of the square shape sample (38 × 38 × 1 cm)

$$\epsilon(f) = B + 2Cf^D + G\{1 - J*(f - H)^2 - j2If\}^{-1}$$

B C D G H I J

0.2974+ 3E-004j	0.6337+ 2E-004j	- 0.3404+ 2E-004j	0.1091+ 4E- 004j	4E-002	4.28E- 002j	-2.6E- 003	
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Polystyrene foam 12.1 lb/ft³ (9–18 GHz)

0.2959+ 1.5E-003j	0.2959+ 1.5E- 003j	1.2E- 003+ 5E-004j	0.2964+ 1.5E- 003j	-1.9E- 003 +1.1E- 003j	-1E-003+ 5E-003j	-2E- 004+ 1E-004j	
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Polystyrene foam 13.9 lb/ft³ (9–18 GHz)

	0.2457+ 1E-004j	0.2923+ 1E-004j	0.1776+ 1E-004j	3.5E- 002 +1E- 004j	8.4E- 002 +1E-004j	-8.62E- 002j	-9.2E- 003
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Polystyrene foam 15.7 lb/ft³ (9–18 GHz)

	0.2454+ 1E-004j	0.2883+ 1E-004j	0.1733	3.31E- 002	8.43E- 002 +1E-004j	8.68E- 002	-9.1E- 003
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Polystyrene foam 17.0 lb/ft³ (9–18 GHz)

	0.3977+ 1E-004j	0.8283+ 5E-004j	- 0.3271+ 3E-004j	3.02E- 002 +2E- 004j	3.47E- 002 +1E-004j	5.48E- 002j	-3.3E- 003
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Polystyrene foam 20.0 lb/ft³ (9–18 GHz)

	0.3381+ 5E-004j	0.3382+ 5E-004j	5.3E- 003+ 2E-004j	0.3386+ 5E-004j	7.3E-003	-1E- 004+ 7.9E- 003j	-8E-004
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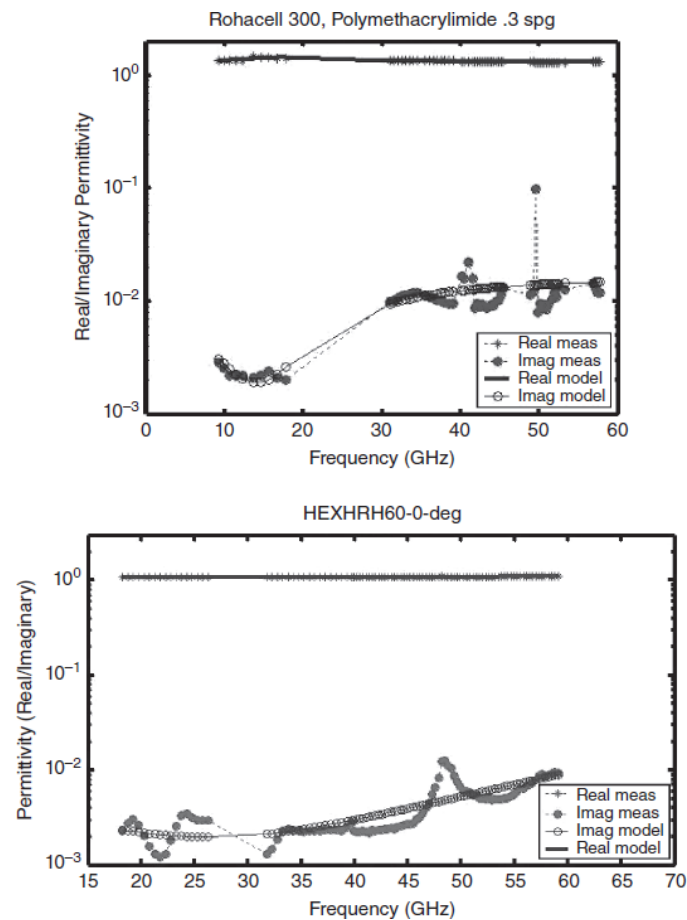
$$\epsilon(f) = B + 2Cf^D + G\{1 - J*(f - H)^2 - j2If\}^{-1}$$

B C D E F G H I J

¼-in. cell, meta-aramid honeycomb, -3-dB/in. transmission loss in X band (200 MHz to 18 GHz)

Sample ID									
0 or 90 degree rotations of the square shape sample (38 × 38 × 1 cm)									
$\epsilon(f) = B + 2Cf^D + G\{1 - J*(f - H)^2 - j2If\}^{-1}$									
	B	C	D	G	H	I	J		
Measurement position 1	1.08+ 0.23j	0.21+ 0.21j	-0.37- 0.34j	0.17+ 0.19j	-0.98- 0.5j	9E-002- 0.31j	2.68+ 0.37j	0.77- 0.11j	-8E-002- 0.13j
Measurement position 2	1.175+ 0.18j	-0.16- 2E-002j	-1.12+ 0.56j	-0.17- 2E-002j	-0.93	1.12- 0.1j	0.35+ 0.81j	0.29+ 2E-002j	-9E-002- 1E-002j
Measurement position 3	1.2+ 0.18j	9E-002+ 0.2j	-1.03- 0.56j	9.4E-002 +0.2j	-0.66- 0.48j	0.12- 0.43j	1.97- 1.04j	0.6- 0.5j	-3E-002- 0.18j

Figure 12.10 Measurement and model fit for a foam (upper) and honeycomb (lower) composite.



Disclaimer: Data are results obtained by the author and colleagues on a small number of samples purchased or obtained as free samples from suppliers. Data are not "guaranteed" by the manufacturer or author. Variations in properties may be observed and users of the products should verify properties. All material properties are guides for engineering design. Materials prepared in the laboratory should be considered as experimental and also guides for engineering design.